

Known Issues, Gaps & TODOs

Critical Hardware & Baud Conflicts

- **Pin Allocation Conflict:** Both `Servo.cpp` and `MotorPWM.cpp` currently attempt to use Pin 9.
- **Baud Rate Mismatch:** Standalone C++ test files open serial at **9600 baud**, while `Nanotransmissionreceiver.cpp` operates at **115200 baud**.

Open Development Items

- **Integration:** Re-enable and wire the commented-out actuator hardware calls in `Nanotransmissionreceiver.cpp` to the servo and motor routines.
- **PID System:** Write the standalone PID steering loop and reference angle calculation modules.
- **Color Bounds:** Replace temporary inline U/V test bounds with dedicated Red and Green calibration sets.
- **Transmission Delay:** Tune the `time.sleep(0.1)` transmission loop delay against real-time camera FPS.
- **Code Cleanup:** Remove test scaffolding (`input()` calls in Python, terminal prints in vision module, test print statements).
- **Optimization:** Evaluate dropping `center_y` calculation if unneeded by the PID avoidance algorithm.